SPACE ENVIRONMENT EFFECTS ON ADCS TELEMETRY

Environmental Context Guide for Anomaly Interpretation
ESA OPS-SAT Mission Operations
Version 1.0

1. THE SPACE ENVIRONMENT AT OPS-SAT ORBIT

OPS-SAT orbited at approximately 515-520 km altitude in a Sun-synchronous orbit (SSO) with inclination ~97.5°. This orbital regime exposes the satellite to several environmental effects that directly produce or mimic ADCS telemetry anomalies. Understanding these effects is essential to distinguish genuine hardware faults from expected environmental responses.

2. EARTH'S MAGNETIC FIELD VARIATION (IGRF MODEL)

The geomagnetic field at OPS-SAT altitude varies systematically with orbital position. The key characteristics operators must know:

Field strength range:
- Equatorial regions: ~25,000 nT (25 ?T)
- Polar regions: ~55,000-60,000 nT (55-60 ?T)
- Average in LEO: ~40,000 nT (40 ?T)

Field direction:
- At any given orbit position, the field vector has a specific orientation in the Earth-fixed frame
- As the satellite moves along its orbit, the field vector rotates through the satellite body frame at approximately one full rotation per orbit

Expected magnetometer behaviour:
- Each magnetometer channel should show a smooth sinusoidal-like variation as the satellite moves through the field
- The period of variation is the orbital period (~95 minutes)
- Rapid variations within a single orbit are anomalous (unless tumbling)
- The International Geomagnetic Reference Field (IGRF) model provides the expected field at any orbital position with ~100 nT accuracy

South Atlantic Anomaly (SAA):
- Located roughly between -50° to +10° latitude, -90° to +30° longitude
- Field strength in this region is 20-30% LOWER than at equivalent latitudes elsewhere
- The weakened field allows more energetic particles to penetrate to lower altitudes
- During SAA passes: single-event upset rates increase significantly
- Magnetometer readings in the SAA may show transient spikes or reduced variance due to genuinely lower field strength

Polar regions:
- Field strength peaks at the magnetic poles (~60 ?T)
- Magnetometer readings naturally reach their maximum values here
- This is NORMAL and should not trigger anomaly alerts in isolation

3. RADIATION ENVIRONMENT EFFECTS

SINGLE-EVENT UPSETS (SEU):

Cause: High-energy cosmic ray or trapped particle strikes the digital circuitry of the ADC or OBC, flipping a bit in a register.

ADCS telemetry manifestation:
- Single-sample spike in one channel: one frame shows an extreme value (often at ADC minimum or maximum) then returns to normal
- Appears as a brief point anomaly
- More frequent during SAA passes
- Can occur in any digital channel

Treatment: Single-frame spikes that return to normal are NOT hardware faults. They are normal SEU events. Apply median filtering to remove them. Do not take corrective action on single-frame spikes alone.

TOTAL IONISING DOSE (TID):
Cause: Accumulated radiation damage to semiconductor materials over time.

ADCS telemetry manifestation:
- Gradual shift in sensor bias (weeks to months)
- Gradual decrease in photodiode sensitivity (the photodiode junction degrades; generates less current per unit of incident light)
- Gradual increase in amplifier offset voltages

Treatment: Regular calibration updates. Monitor trend lines over mission lifetime. Update calibration tables quarterly.

DISPLACEMENT DAMAGE:
Cause: High-energy particles displace atoms in semiconductor crystal lattice, creating permanent defects.

ADCS telemetry manifestation:
- Increased dark current in photodiodes (non-zero reading in darkness)
- Reduced photosensitivity
- Increased thermal noise floor in all analog sensors

4. THERMAL ENVIRONMENT EFFECTS

OPS-SAT experiences a large thermal cycle with each orbit:
- Sunlit phase: temperature range +20°C to +60°C for external surfaces
- Eclipse phase: temperature range -20°C to -40°C for external surfaces
- Cycle period: ~95 minutes

Thermal effects on ADCS telemetry:

MAGNETOMETER THERMAL DEPENDENCE:
- Fluxgate magnetometers have temperature-dependent sensitivity (typical: ±0.1% per °C)
- Temperature-dependent offset (typical: ±0.1 ?T per °C)
- At OPS-SAT thermal swing of ~80°C, this can produce ±8 ?T bias variation
- The bias follows the thermal cycle (period = orbital period)

Identification: Magnetometer bias that correlates with eclipse/sunlit transitions and varies with ~95-minute period is thermal in origin.
Corrective action: Apply temperature-compensated calibration tables.

PHOTODIODE THERMAL DEPENDENCE:
- Photodiode dark current increases exponentially with temperature
- At high temperatures (>40°C), dark current may produce non-zero

readings even in darkness (eclipse dark current anomaly)
- Sensitivity also varies with temperature

Identification: Non-zero readings during eclipse that increase in the second half of the sunlit phase (when satellite is warmest). Returns to zero as satellite cools in eclipse.
Corrective action: Apply temperature correction coefficient or increase dark-reading threshold.

ELECTRONICS THERMAL EFFECTS:
- ADC reference voltages are temperature-dependent
- At extreme temperatures, analog circuits may malfunction
- Very cold temperatures (<-30°C) can cause temporary ADC failures

Monitoring: ADCS electronics temperature telemetry should be monitored alongside sensor telemetry.

5. ECLIPSE AND ECLIPSE TRANSITION EFFECTS

ECLIPSE ENTRY (Day-to-Night transition):
Expected sequence:
1. Photodiodes: gradual darkening over ~5-10 seconds as satellite enters penumbra, then rapid drop to near-zero in umbra
2. Power subsystem: switches from solar charging to battery discharge
3. Temperature: begins to drop for external components
4. ADCS: switches to magnetometer-only attitude determination
5. Magnetorquers: may command manoeuvre to maintain sun-pointing for eclipse exit preparation

Anomaly markers:
- Photodiodes fail to darken at predicted eclipse entry ? attitude wrong or ephemeris error
- Magnetometers show unusual noise at eclipse entry ? may be caused by EPS mode change generating transient EMI

ECLIPSE EXIT (Night-to-Day transition):
Expected sequence:
1. Photodiodes: gradual brightening starting with Sun-facing faces
2. Power subsystem: transitions from battery to solar panel charging
3. Temperature: begins to increase
4. ADCS: can now use photodiodes again for sun vector estimation

Anomaly markers:
- Photodiodes do not brighten at predicted eclipse exit ? tumbling (faces not pointing toward Sun) or failed photodiodes
- Asymmetric brightening (only some faces) ? may indicate attitude drift

6. SUN-RELATED ENVIRONMENTAL EFFECTS

SUN ILLUMINATION GEOMETRY:
The angle between the Sun and each satellite face determines photodiode output according to cosine law:
$V_{out} = V_{max} \times \cos(?)$
where ? is the angle between the face normal and the Sun direction.

Key geometric scenarios:
- Face directly Sun-pointing (?=0°): maximum output (may saturate)
- Face at 90° to Sun: near-zero output (edge-on illumination)

- Face pointing away from Sun: zero output
- Multiple faces partially illuminated: this is normal during transitional attitudes

SOLAR EVENTS:
- Solar flares can temporarily increase solar irradiance by 1-10%
- This may cause slight increase in photodiode readings (easily missed)
- Solar proton events following major flares increase SEU rates dramatically
- During X-class solar flares: expect elevated single-event upsets for 24-72 hours
- Monitor NOAA SWPC (Space Weather Prediction Center) for solar events

EARTH ALBEDO:
- Earth reflects approximately 30% of incident solar radiation (albedo)
- The Earth-shine component can partially illuminate the nadir-pointing face of the satellite even when not directly Sun-pointing
- This creates anomalously bright readings on the nadir photodiode that can complicate sun vector determination
- Most significant at low orbital altitudes and over high-albedo regions (ice, clouds, desert)

Identification: Nadir-pointing photodiode reading 10-30% of maximum intensity when the Sun is far from nadir. This is normal albedo input.

7. ELECTROMAGNETIC ENVIRONMENT

OPS-SAT INTERNAL EMI SOURCES:
Primary sources affecting ADCS measurements:

1. MAGNETORQUERS: Generate magnetic fields during operation.
   Affected channels: CADC0872, CADC0873, CADC0874 (all magnetometers)
   Effect: High noise during operation; residual field after deactivation (decay time: 5-30 minutes depending on residual magnetisation)

2. ELECTRICAL POWER SYSTEM (EPS): Battery charge/discharge switching.
   Affected channels: All analog channels potentially
   Effect: Transient spikes at switching frequencies

3. REACTION WHEELS: Electric motors generate magnetic fields.
   Affected channels: Magnetometers, primarily
   Effect: Steady-state bias proportional to wheel speed; vibration

4. ON-BOARD COMPUTER: Clock signals, memory access
   Affected channels: Typically low; may affect ADC if grounding poor
   Effect: Regular periodic noise at OBC clock frequency or harmonics

5. EXPERIMENT PAYLOADS: OPS-SAT hosted various experiments.
   Affected channels: Highly variable depending on experiment
   Effect: May generate EMI unpredictably; must correlate with payload activity logs

SHIELDING LIMITATIONS:
CubeSat platforms have limited volume for electromagnetic shielding. The compact form factor means that magnetometer placement is compromised ? magnetometers are near other electronic components. This makes OPS-SAT particularly susceptible to internal EMI compared to larger spacecraft.

8. ORBITAL GEOMETRY EFFECTS ON SPECIFIC CHANNELS

CADC0872 (Magnetometer X):
  Peaks and troughs: Twice per orbit as orbital trajectory crosses
  field lines. Peaks near magnetic equator crossings; troughs in
  between for certain orbit inclinations.

CADC0873 (Magnetometer Y):
  Similar to X but ~90° phase shifted. The Y and X components should
  show a characteristic phase relationship if the satellite is properly
  oriented.

CADC0874 (Magnetometer Z):
  For an SSO orbit, the Z (nadir) component shows less variation than
  X and Y over one orbit. Unusually large variation in Z suggests
  attitude deviation from nominal nadir pointing.

CADC0875 (+X Photodiode):
  Illuminated when the +X face points toward the Sun. For OPS-SAT in its
  nominal attitude, this follows a predictable daily and orbital pattern.

CADC0876 through CADC0880 (remaining photodiodes):
  Each follows its own orbital illumination pattern depending on the
  satellite's nominal attitude mode and sun angle.

9. CROSS-REFERENCING WITH EXTERNAL DATABASES
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When diagnosing ADCS anomalies, operators should consult:

1. NOAA SPACE WEATHER PREDICTION CENTER (swpc.noaa.gov):
   - Real-time Kp index for geomagnetic activity
   - Solar flare alerts and X-ray flux measurements
   - Proton event forecasts (SEU risk indicator)

2. WORLD MAGNETIC MODEL (WMM):
   - Alternative to IGRF for geomagnetic field predictions
   - Updated every 5 years by NOAA/BGS
   - Available at ngdc.noaa.gov/geomag/WMM

3. CELESTRAK / SPACE-TRACK (celestrak.com / space-track.org):
   - OPS-SAT orbital elements (TLE) for eclipse prediction
   - Eclipse/sunlit calculations for any given time and orbit

4. NASA OMNIWEB (omniweb.gsfc.nasa.gov):
   - Historical space weather data
   - Radiation environment history for OPS-SAT orbit

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END OF SPACE ENVIRONMENT EFFECTS GUIDE
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